

Engineering Portfolio

joelbhattarai.com/projects.html



Executing program on UR3e-540

Universal Robots Programming

Designed program that makes arm draw a square, circle, and triangle, without "picking up the pencil", all with only one hard coded position or "Waypoint", and making larger and larger shapes about the same center point.

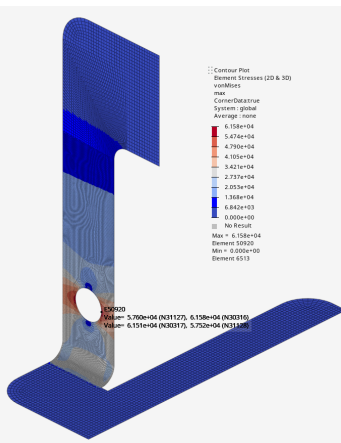


*Testing controller on Quanser
Qube-Servo 3*

Controls

Performed design of PID controllers in time and frequency domain. Designed full-state feedback and compensated full-state feedback controllers. Pictured full-state controller was designed to stabilize an inverted pendulum system with a full-state feedback controller built on Simulink.

Details/Documentation

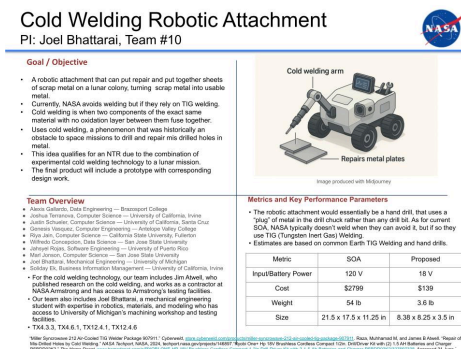


Von Mises stress contour on a bookstop

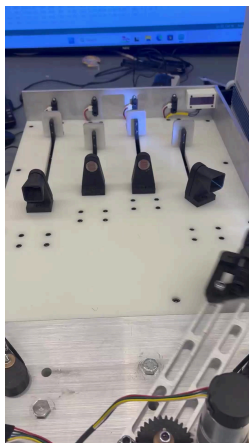
Finite Element Analysis

Analyzed a variety of shapes in HyperMesh and OptiStruct. Leveraging 2D and 3D elements, convergence studies, and buckling analysis, to make design recommendations on a variety of systems.

Details/Documentation



*Quad chart for cold welding
attachment concept*



Linkage playing "game" to shine light on four different targets based on block positions.

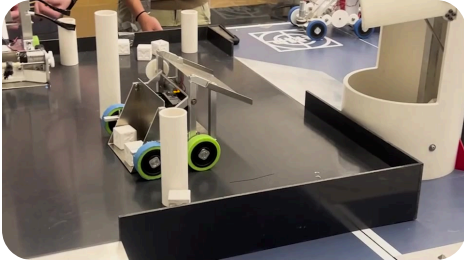
Principal Investigator

Proposed an new technology concept that was selected by a team of NASA engineers and stakeholders. Led a team of interdisciplinary students to a full technical proposal including a NASA New Technology Report. Also served on a NASA review board for three other proposals as the primary reviewer on the panel.

Linkage Development

Leveraged MSC Adams for multibody dynamic simulation to model the linkage system to hit all four targets. Designed transmission, links, and hardstops for a robust solution. Performed PID tuning to ensure the speed of performance and fast settling.

Details/Documentation

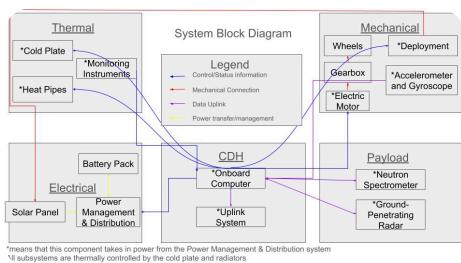


Robotic Machine Player confirming points for our team by delivering cubes into the delivery location

RMP Development

Developed CAD and design skills to complete multiple different tasks in a high-pressure competition. Designed robust transmission and powertrain systems. Executed analytical statics analysis to ensure the functionality of multiple functions simultaneously.

Details/Documentation



Full rover system block diagram

Lead Systems Engineer

Led a group of engineering students to a PDR for a lunar rover mission. Coordinated with the Science and Programmatic team on developing system-level requirements, and flowed these requirements down to Mechanical, Electrical, and Thermal requirements. Managed all interfacing between systems as well as protocol development.